

SocializeChat: a GPT-based AAC Tool for Social Communication Through Eye Gazing

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ABSTRACT

People with motor and speech impairments has limited communication abilities, resulting in a heavy reliance on Augmentative and Alternative Communication (AAC) tools. Existing commercial AAC tools provide simple combination of fixed words, satisfying basic physiological needs while encounter huge challenges in social communication. This social communication is significant for users' mental health, especially for those with limited motor abilities. This article thus developed SocializeChat, an AAC tool that employs LLM technology to boost social chat with gaze inputs. Specifically, SocializeChat generates multiple sentences of conversation in real time, offers suggestions tailoring to users' preferences of topics, and phrases sentences in a style in accord with the relationship of people in conversation. This is achieved through a user dataset containing content preferences and social closeness, and through dedicated design of prompts and procedures. In a brief testing, SocializeChat was rated as effectively embodying subjects' own content preferences and communication style.

CCS CONCEPTS

• **Human-centered computing**; • **Ubiquitous and mobile computing**; • **Ubiquitous and mobile computing systems and tools**;

KEYWORDS

Augmentative and Alternative Communication (AAC), Large Language Models (LLM), Social Communication, Content Preferences, Social Closeness

ACM Reference Format:

Yuyang, Fang, Yunkai, Xu, Zhuyu, Teng, Zhaoqu, Jiang, and Wei, Xiang. 2023. SocializeChat: a GPT-based AAC Tool for Social Communication Through Eye Gazing. In *Adjunct Proceedings of the 2023 ACM International Joint*

Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct), October 08–12, 2023, Cancun, Quintana Roo, Mexico. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3594739.3610705>

1 INTRODUCTION

Diseases such as stroke, Parkinson's, and Amyotrophic lateral sclerosis (ALS) may lead to severe impairments in the motor and speech abilities of patients. Individuals with such impairments relying on Augmentative and Alternative Communication (AAC) for communication, sometimes solely through gaze input, as shown in Figure 1. Currently, AAC tools support speech production while seldom social interaction [1], resulting in social isolation [2], and psychological health issues [3, 4]. Enriching the social communicative abilities of AAC users holds significant importance in alleviating social isolation and improving psychological well-being [5, 6].

Recently, AAC applications become the mainstream tool with the advancements in edge computing and mobile applications [7]. These tools have made significant progress, but have not performed well in social communication because of the following reasons: (1) Limited and impersonal content: AAC tools offer fixed vocabulary and short sentences, which seldom change with AAC users' daily lives. (2) Static and predefined styles: AAC tools lacking the ability to adapt communication style with the conversation partner's relationship [8]. Although recent studies have integrated Large Language Models into AAC and produced sentences according to simple inputs [9–11], they still emphasized AAC users' input speed rather than content optimization.

This article design SocializeChat based on LLM, aiming to assist AAC users in their social communication. SocializeChat utilized GPT-4 to describe two features during communication, content preferences [12, 13], and social closeness [14], forming a descriptive dataset of users. Based on the dataset, the LLM generates content that matches user's preferences and communication style with people of varied relationships. The interface of SocializeChat was designed in the form of eye-gaze input. An use case displayed promising potential of SocializeChat in addressing the social participation challenges faced by individuals with motor and speech impairments.

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UbiComp/ISWC '23 Adjunct, October 08–12, 2023, Cancun, Quintana Roo, Mexico

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ACM ISBN 979-8-4007-0200-6/23/10.

<https://doi.org/10.1145/3594739.3610705>



Figure 1: An AAC User with Motor and Speech Impairments

2 RELATED WORK

Researchers used LLMs in AAC tools to help users regain communication abilities. The main methods focused on entry rate, aiming to help keyboard-based AAC users have the same text entry rate as able-bodied people. Cai used LLMs to generate complete sentences from input abbreviations and conversational context [11], achieving an effective keystroke savings rate of approximately 77%. Kwickchat supported word prediction and generated multiple alternative options [9], resulting in a 65% reduction in keystroke costs and being perceived by users as having good response quality. These studies have accelerated entry rates for AAC users to varying degrees, reducing the rate gap in communication between them and able-bodied people. However, few studies optimized the content and style of communication, leaving social communication an unsolved problem.

With the rapid developing of LLM capabilities, it is important to explore high-level communication goals such as social communication [8]. Dai et al. proposed a direction that used AAC to support relationship maintenance and social communication [8]. Valencia used LLM-designed speech macros to extend reply with background information [10]. Multiple reply suggestions were provided by speech macros, allowing users to choose based on the conversation partner's identity, mood, and circumstance, to achieve a certain degree of response diversity [8]. However, various content preferences of AAC users have not been differentiated yet [8], and the generated sentences by LLMs are abrupt and straightforward in style [10], affecting their social relationships with conversation partners.

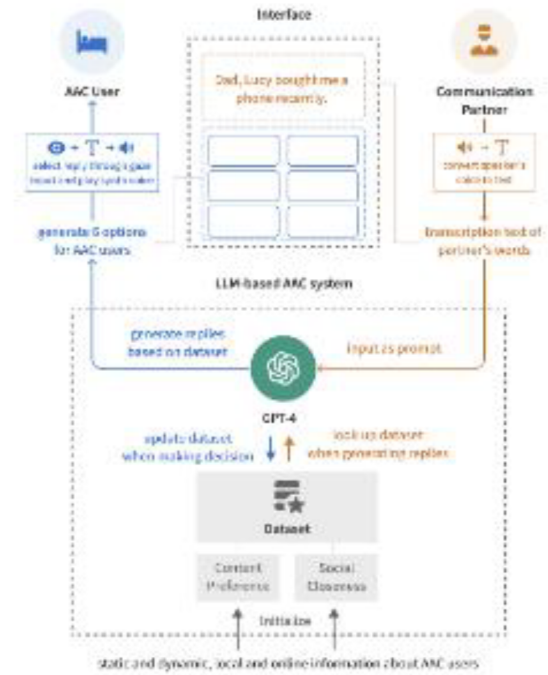


Figure 2: SocializeChat, an LLM-based AAC tool

3 SYSTEM ARCHITECTURE

3.1 LLM-based AAC Tool

SocializeChat is an AAC application with eye-gaze input, supporting the social communication of users with motor and speech impairments. The system is based on the conversational scene as seen in Figure 2. Whenever users want to start a conversation and reply others' chats and queries, they can turn to SocializeChat and choose what they tend to say. Then, SocializeChat synthesizes speeches to chat with others.

Like other researchers using background information [9] and conversational context [11], SocializeChat also has a dataset involving content preferences and social closeness in order to provide prompt reference for GPT-4, as seen in Figure 2. It employs GPT-4 to generate sentence suggestions that are consistent with AAC user's social relationships, living status, and personal habits using these two dimensions. Specifically, user's preferences have an impact on the sentence suggestions generated by GPT-4, while social closeness affects communication style.

The application begins with both static and dynamic, local, and online information about AAC users (e.g., personal blogs, diaries, frequently viewed websites and newspapers, etc.) and is initialized when GPT-4 extracts which topic they prefer and how much they are interested in it. It also utilizes GPT-4's internet browsing and inference capabilities. As AAC users continuously make decisions about certain generated sentences, the dataset will be continuously updated and improved, making sentences in line with users' personal wishes.

There are two notable features of this tool:

1. Updatable Local Dataset

Fixed information leads to relatively repetitive topics and a diminished communication experience. We employ the Internet browsing and inference capabilities of GPT-4 in our system to update dataset. There are three sources for the system's data: the preset information inputted during the initial use of AAC (such as personal information, family social relationships, diaries, blogs, etc.), online information relevant to the interests of the users obtained through GPT-4's Internet browsing capabilities, and information continuously supplemented during the conversation. The dataset is stored in JSON format and consists of two objects. The "Conversation Topics" object contains three key values: "Topic", "Content Preferences Level" and "Reason". The JSON array object "Social Closeness" contains objects with different names and levels of closeness, with attributes such as "Name" and "Closeness Level".

2. Basis for Adjusting Generated Suggestion

To optimize sentence suggestions that align with users' social communication intents, we prioritize the findings regarding 'content preference' and 'social closeness'. These two factors are derived from our considerations of dialogue semantic content and linguistic style, aiming for a clear understanding of the users. Notably, the selection of these two elements considered the users' preferences towards multiple topics [8]. Additionally, from interviews with family members of person with severe motor and speech impairments, we found these two elements to be most directly relevant to social conversation. This interview involved three caregivers and all of them are the person's close relatives.

"My husband has difficulty speaking due to a nasal feeding tube. When I take care of him, he always makes noises to keep me a little longer, but not with professional nurses." – Caregiver 1, wife of the person

"Due to work reasons, I didn't meet my dad very often when I grew up. I really want to know the story of his youth and hope he can give me some life experience." – Caregiver 2, daughter of the person

"Although we knew he couldn't join our conversation, we always shared recent events and news in front of him, hoping that he would also have a better understanding of what was going on around him. He always sat on the bed and listened quietly, and we didn't know what was on his mind." – Caregiver 3, granddaughter of the person

Also, in some previous user studies [9, 10], the AAC tool can perform better in the conversation if the researchers consider these two factors. Considering that the target users may be motor and speech impaired, an eye-gazing interface is built and serves as the input method for the system.

3.2 Procedure

The whole procedure can be divided into two main phases: initialization phase and communication phase, as seen in Figure 3.

3.2.1 Initialization Phase. Establishment of content preferences and social closeness by GPT-4. In this phase, we provide a defined description of content preferences and social closeness



Figure 3: Procedure of GPT-4 generating sentence suggestions

based on a literature review, which GPT-4 further refines into interpretable levels. The content preferences are set at four levels: enthusiastic engagement, friendly involvement, indifferent response, and distant resistance. Social closeness levels are categorized into three levels: formal, moderate, and close.

The prompts we use to generate levels of content preferences are as follows:

"I envision a hierarchy of positive response levels for topics as follows:

Level 1 - Intimate and interested. It means ···· Level 4 - Distant and disgusted ····

Please refine the current expressions, generate two intermediate levels, and provide the level divisions and reasons."

And the results are as follows:

"Level 1 - Enthusiastic Engagement: ···· Highly interested and ····

Level 2 - Friendly Involvement: ···· Some level of attention ····

Level 3 - Indifferent Response: ···· Not very interested ····

Level 4 - Distant Resistance: ···· Strongly uninterested or disgusted ····"

The prompts we use to generate levels of social closeness are as follows:

"In terms of language style, I divide social closeness into the following three levels:

1. Formal Closeness: Formal closeness is suitable for formal occasions or language styles used when communicating with unfamiliar individuals. ····

·····

Please understand these three levels."

And the results are as follows:

"1. Formal Closeness: This is a conservative language style used for formal occasions or conversations with unfamiliar individuals. ···· The expression tends to be more formal, avoiding overly familiar or casual remarks.

2. Moderate Closeness: ···· Emphasizing friendliness, relaxation, and increased interaction. This style allows

for more casual and relaxed expressions, while still considering the other person's feelings.

3. High Closeness: This style of language is freer and more intimate, possibly including catchphrases, private jokes, and special nicknames, expressing a deep sense of closeness, and understanding."

Topic generation across various types of information. With the Internet browsing capability of GPT-4, we use inputs such as diaries, blogs, and other local information, as well as online information from websites and publications. These inputs are organized into topics and potential questions.

Establishment of the user's dataset. Based on the predefined levels of content preferences and social closeness, GPT-4 assesses the levels of preferences for the topics. It provides a rating for the preference level of the user's responses to different topics, along with the reasoning behind such evaluations. Using the social closeness, GPT-4 determines suitable communication styles for the user based on their relationships with people they talk to. For instance, conversations with strangers may require a more formal tone, while interactions with family members can be more intimate and casual. By combining these two aspects, a comprehensive user's dataset is constructed.

3.2.2 Communication Phase. Analysis of conversation with the conversation partner. A speech-to-text engine (such as Microsoft Azure AI) converts audio information from the partner into text, and GPT-4 extracts content, and determines the level of social closeness. Then, suggestions during the subsequent conversations are based on the same level of closeness.

Matching and generating sentence suggestions. The system incorporates the user's dataset, matching the extracted content with their respective levels of preferences, and generating six sentence suggestions based on the four level content preferences pertaining to the topic. Three of six are in the same preference level that is best matches the user's preference. They mean the same but are randomly generated, aiming to provide more choices.

Refinement of the user's dataset. Ultimately, the system presents these generated sentence suggestions on the frontend as selectable expressions for individuals with such disabilities. Based on the user's selected suggestion, the system will assess the updated content preferences, and the dataset is revised to improve accuracy.

4 PRELIMINARY USE CASE

Figure 4 shows the key functionalities of the main interface. Above is a transcription text of partner's words, and below are 6 options generated for AAC users.

Based on the prototype we developed, we selected three able-bodied people for prototype testing as shown in Figure 4. Three subjects believed that the system was helpful during the conversation around the topic, but the performance was not ideal when they talked aimlessly and casually. In these simple use cases, we provided three topics, including "travel experience", "daily work", "personal hobbies". A round we define is that the user selects one suggestion and another speak to reply. The average conversation rounds around all three topics exceeded 12 in 3 minutes. Here is an example of a participant engaging in a conversation using the

system, with the conversation revolving around the topic of "travel experiences".

The communication partner says:

"Auntie, how is your recent bronchitis? There are so many flowers in the People's Park in spring! I hope you get better soon, and we can go for an outing together."

The GPT-4 extract the topic as "health status, outdoor activities" and generates sentence as follows and the user choose the second suggestion of level 1:

Level1:1-"Going for a spring outing is my favorite activity!" 2-"The springtime in the park, I really want to experience it myself!" 3-"Going for a spring outing in the park sounds nice."

Level2: "Going for a spring outing in the park sounds good."

Level3: "Understood, thank you."

Level4: "I don't want to talk about it right now."

Then the partner continues:

"I'll show you some of the pictures I took with my camera."

The GPT-4 extract the topic as "see the pictures" and generates sentence as follows and the user choose the first suggestion of level 2:

Level 1: "I'm really looking forward to seeing your photos!"

Level 2: 1-"Sure, I'd love to take a look." 2-"I'm really interested in seeing your photos." 3-"Let me see the photos you took."

Level 3: "Want to see the photos? Sure."

Level 4: "I'm not really in the mood to look at photos right now."



Figure 4: A User imitating the using context of SocializeChat

When we did not specify a topic for the conversation, the test users reported that the options were not ideal. Without a specific topic, the system of our tool might not correctly determine users' preference. Therefore, it would generate suggestions with slight difference between the four levels, making our users confused. Some potential improvements, such as an independent mode for conversations without certain topic, may enable our tools to be used in complex social situations. However, more user testing is required in the future to validate our initial analysis.

5 CONCLUSION AND FUTURE WORK

This article proposes SocializeChat, an LLM-based AAC system that generates tailored sentence suggestions based on user's content preferences and social closeness to their partner. It employs GPT-4 to achieve multi-turn conversations, and follows the widely used eye-gaze input to support a broad range of impaired users. Then, a preliminary test was conducted.

There are still many things to do in the future. In the initialization phase, other input forms can be used in addition to text, like images and recorded sound, to better understand AAC users. In the communication phase, the updating and improving process of content preferences and social closeness still needs testing for its stability. The availability of GPT-4 tokens is restricted, leading to limited turns in actual conversation. Also, culture is a significant aspect of AAC users [15] and needs to be taken in consideration. Users of our AAC tool may have different culture background, which may affect their content and communication style. Meanwhile, types of closeness, which are determined by the relationship between users and speakers, may also have influence on the content.

In addition, more tests of the target user need to be conducted. The potential users of our system may have unexpected demands for social communication. Therefore, we have planned to evaluate the usability of our system and the actual help for social communication scenarios.

We do hope that SocializeChat can provide some inspiration for further research on the integration of AAC and LLM, and pave the way for them to explore the ability to empower those with motor and speech disabilities for social communication.

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